nic_demo

```r
knitr::opts_chunk$set(echo = TRUE, message = FALSE, warning = FALSE)
```

Introduction

My attempt to understand and reproduce the work done in the synthpop repo by j4sonli.

Set up Environment

^ Only need to run once

```r
# 1. Initialise a renv in the project directory, you can check
# it's created in your local machine by running `.libPaths()`
# renv::init()

# 2. Install the packages j4sonli recommended in his README
# renv::install(c(c("dplyr","tidyr","stringr","R.utils","tidycensus","bnlearn","foreach","doSNOW","ggpl

# 3. Update your lockfile
# renv::snapshot()
```

Load Libraries

```r
library(glue)
library(config)
library(dplyr)
library(ggplot2)
```

```r
# Test getting api key from config file
# Install if necessary: install.packages("config")
cfg <- config::get()
us_census_api_key <- cfg$us_census_api_key
```

```r
STATE_FIPS <- data.frame(
  state_name = c(
    "Alabama", "Alaska", "Arizona", "Arkansas", "California", "Colorado",
    "Connecticut", "Delaware", "District of Columbia", "Florida", "Georgia",
    "Hawaii", "Idaho", "Illinois", "Indiana", "Iowa", "Kansas", "Kentucky",
    "Louisiana", "Maine", "Maryland", "Massachusetts", "Michigan", "Minnesota",
    "Mississippi", "Missouri", "Montana", "Nebraska", "Nevada", "New Hampshire",
```

```r
    "New Jersey", "New Mexico", "New York", "North Carolina", "North Dakota",
    "Ohio", "Oklahoma", "Oregon", "Pennsylvania", "Rhode Island", "South Carolina",
    "South Dakota", "Tennessee", "Texas", "Utah", "Vermont", "Virginia",
    "Washington", "West Virginia", "Wisconsin", "Wyoming"
  ),
  state_abb = c(
    "AL", "AK", "AZ", "AR", "CA", "CO", "CT", "DE", "DC", "FL", "GA",
    "HI", "ID", "IL", "IN", "IA", "KS", "KY", "LA", "ME", "MD", "MA",
    "MI", "MN", "MS", "MO", "MT", "NE", "NV", "NH", "NJ", "NM", "NY",
    "NC", "ND", "OH", "OK", "OR", "PA", "RI", "SC", "SD", "TN", "TX",
    "UT", "VT", "VA", "WA", "WV", "WI", "WY"
  ),
  state_code = c(
    "01", "02", "04", "05", "06", "08", "09", "10", "11", "12", "13",
    "15", "16", "17", "18", "19", "20", "21", "22", "23", "24", "25",
    "26", "27", "28", "29", "30", "31", "32", "33", "34", "35", "36",
    "37", "38", "39", "40", "41", "42", "44", "45", "46", "47", "48",
    "49", "50", "51", "53", "54", "55", "56"
  ),
  stringsAsFactors = FALSE
)

head(STATE_FIPS)
```

```
##   state_name state_abb state_code
## 1    Alabama        AL         01
## 2     Alaska        AK         02
## 3    Arizona        AZ         04
## 4   Arkansas        AR         05
## 5 California        CA         06
## 6   Colorado        CO         08
```

Cambridge Massachusetts

Sample synth pop dataset in repo

```r
output_cambridge_hhs <- read.csv("./synthpop_output/2503302/syn_hhs_spatial_2503302-example.csv")

head(output_cambridge_hhs)
```

```
##   HHID                       tenur   hhinc nwork ncars nchil hhsiz true_hhsiz
## 1    1 Owned with mortgage or loan   >150K     2    2+     0     2          2
## 2    2                      Rented   1-25K     0     1     0     2          2
## 3    3                      Rented  51-75K     2     1     0     2          2
## 4    4                      Rented   >150K     2     1     1     3          3
## 5    5                      Rented  51-75K     2    2+     1     4          4
## 6    6                      Rented 76-100K     1     0     1     3          3
##   true_nwork true_nchil hhtype    puma i_age i_sex race_ i_eth i_inc emply
## 1          2          0     MC 2503302  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 2          0          0     MC 2503302  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 3          2          0     MC 2503302  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
```

```
## 4          2          1     MC 2503302  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 5          2          1     MC 2503302  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 6          1          1     MC 2503302  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
##   indus edu_q tenur_prox        geoid
## 1    NA  <NA>      Owned 250250706002
## 2    NA  <NA>     Rented 250250702001
## 3    NA  <NA>     Rented 250250107022
## 4    NA  <NA>     Rented 250250305002
## 5    NA  <NA>     Rented 250250510002
## 6    NA  <NA>     Rented 250250708001
```

Fulton County, Georgia

```
# Federal Information Processing Standard (Federal Information Processing Standard), two-digit numeric
PUMA_CODE <- GA_puma_codes[1]
GA_STATE_CODE <- STATE_FIPS[STATE_FIPS$state_name == STATE_NAME,]$state_code
run_command <- glue('Rscript synthpop.R -puma={GA_STATE_CODE}00{PUMA_CODE}')

# Run the following command in your terminal to generate a synthpop of Fulton County, where Atlanta is
print(run_command)
```

```
## Rscript synthpop.R -puma=1300100
```

Output

```
# Where Atlanta is located
output_fulton_hhs <- read.csv("./synthpop_output/1300100/syn_hhs_1300100.csv")
head(output_fulton_hhs, 5)
```

```
##   HHID                      tenur    hhinc nwork ncars nchil hhsiz true_hhsiz
## 1    1         Owned free and clear   51-75K     0    2+     0     2          2
## 2    2 Owned with mortgage or loan   51-75K     1    2+     0     3          3
## 3    3                     Rented 101-150K     2    2+     0     3          3
## 4    4         Owned free and clear   51-75K     0    2+     0     2          2
## 5    5 Owned with mortgage or loan 101-150K     1    2+     0     3          3
##   true_nwork true_nchil hhtype    puma i_age i_sex race_ i_eth i_inc emply
## 1          0          0     MC 1300100  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 2          1          0     MC 1300100  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 3          2          0     MC 1300100  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 4          0          0     MC 1300100  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 5          1          0     MC 1300100  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
##   indus edu_q tenur_hhinc_prox tenur_hhsiz_prox tenur_prox
## 1    NA  <NA>     Owned.51-75K          Owned.2      Owned
## 2    NA  <NA>     Owned.51-75K          Owned.3      Owned
## 3    NA  <NA>  Rented.101-150K         Rented.3     Rented
## 4    NA  <NA>     Owned.51-75K          Owned.2      Owned
## 5    NA  <NA>   Owned.101-150K          Owned.3      Owned
```

Comparison (Cambridge v. Fulton)

```
# Intersect columns from
intersected_colnames <- intersect(colnames(output_cambridge_hhs), colnames(output_fulton_hhs))

for (colnames in intersected_colnames) {
  print(colnames)
}
```

```
## [1] "HHID"
## [1] "tenur"
## [1] "hhinc"
## [1] "nwork"
## [1] "ncars"
## [1] "nchil"
## [1] "hhsiz"
## [1] "true_hhsiz"
## [1] "true_nwork"
## [1] "true_nchil"
## [1] "hhtype"
## [1] "puma"
## [1] "i_age"
## [1] "i_sex"
## [1] "race_"
## [1] "i_eth"
## [1] "i_inc"
## [1] "emply"
## [1] "indus"
## [1] "edu_q"
## [1] "tenur_prox"
```

Maps

Create directories

to store census information. Only run if you don't have the expected data directories.

```
# base_path <- "synthpop_data/acs_marginals"
#
# org_map_pumas <- c("0603701", "0603716", "0603733",
#                 "4805000", "4804901", "4804614",
#                 "0400134", "0400105", "0400122",
#                 "2501400", "2503603", "2503302",
#                 "1208900", "1203105", "1203103",
#                 "5600200", "5600100", "5600300", "5600400", "5600500")
#
#
# for (map_puma in org_map_pumas) {
#   # Dynamically construct the path: "synthpop_data/acs_marginals/0603701"
#   full_puma_dir <- file.path(base_path, map_puma)
#
```

```
#   if (!dir.exists(full_puma_dir)) {
#     dir.create(full_puma_dir, recursive = TRUE, showWarnings = FALSE)
#     cat("Initialized folder:", full_puma_dir, "\n")
#   }
# }
```

Download Census Data For Demo

Only run again if you don't have the data yet.

```
# org_map_pumas <- c("0603701", "0603716", "0603733",
#                  "4805000", "4804901", "4804614",
#                  "0400134", "0400105", "0400122",
#                  "2501400", "2503603", "2503302",
#                  "1208900", "1203105", "1203103",
#                  "5600200", "5600100", "5600300", "5600400", "5600500")
# for (org_puma in org_map_pumas) {
#   command <- glue('Rscript synthpop_truncated.R -puma={org_puma}')
#   print(command)
# }
```

Import `plot_maps`

```
box::use(./plot_maps)
```

List of functions in plot_maps

- ☒ plot_study_areas()
- ☒ plot_error_heatmaps()
- ☒ plot_error_map()
- ☐ plot_tract_hhtype_diffs_map()
- ☐ plot_pie()
- [X / Helper] error_heatmap()
- [Helper] rmse()
- [Helper] get_rmses()
- [Helper] process_marginals()
- [Helper] get_error_map_data()

Constants

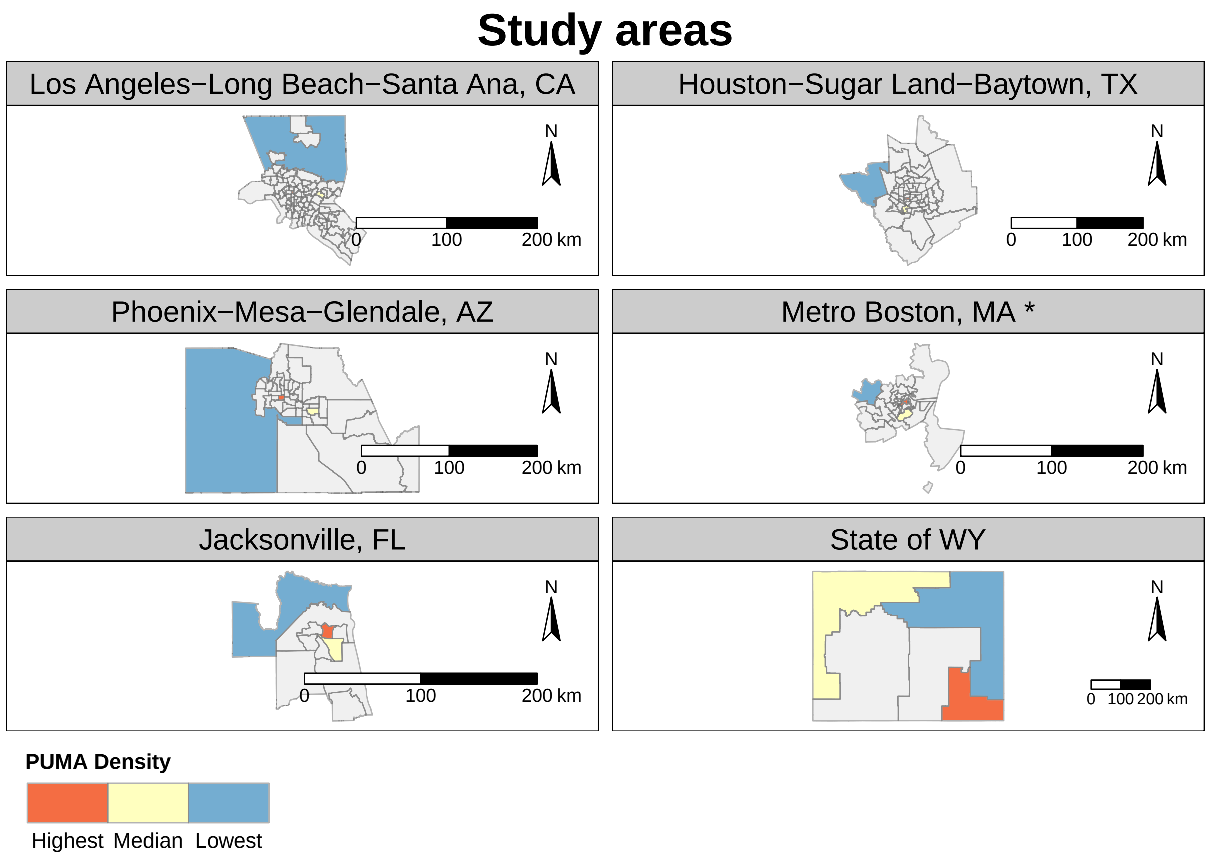
```
map_pumas <- c("0603701", "0603716", "0603733",
               "4805000", "4804901", "4804614",
               "0400134", "0400105", "0400122",
               "2501400", "2503603", "2503302",
               "1208900", "1203105", "1203103",
               "5600200", "5600100", "5600300", "5600400", "5600500")

densities <- c(383.7, 9049.6, 42788.1,
```

```
                66.1, 3457.2, 11736.8,
                112.0, 3947.0, 8075.6,
                885.9, 5817.3, 28710.4,
                160.5, 1989.8, 3285.1,
                # last entry is density for entire state of WY
                5.97)
synpop_suffix <- "20220127"
```

Trial Plot Functions

```
# plot_study_areas()
box::reload(plot_maps)
plot_maps$plot_study_areas(map_pumas, save_pdf=FALSE)
```

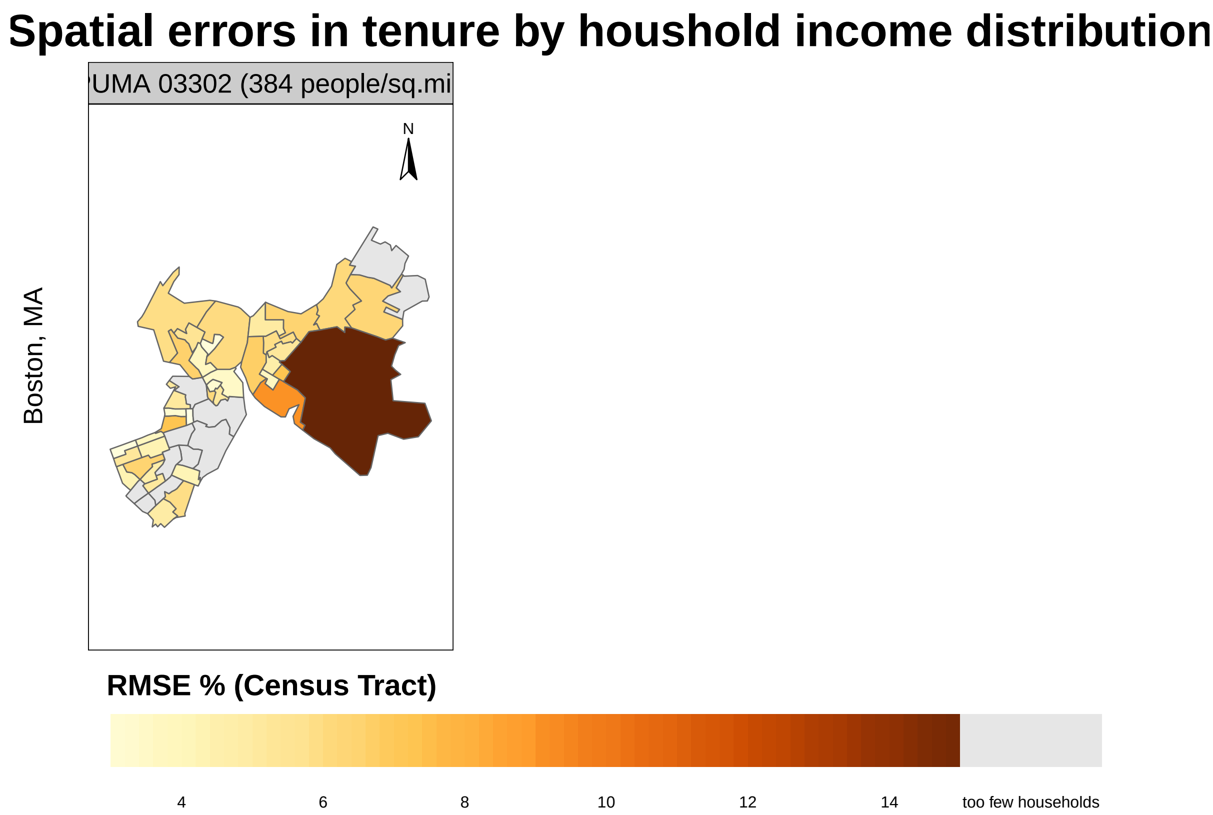
```
## Reading layer 'tl_2010_us_cbsa10' from data source
##   'C:\Users\nicho\Documents\git\us-synth-pop\synthpop\map_data\tl_2010_us_cbsa10\tl_2010_us_cbsa10.sl
##   using driver 'ESRI Shapefile'
## Simple feature collection with 955 features and 12 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: -178.4436 ymin: 17.83151 xmax: -65.42271 ymax: 65.45448
## Geodetic CRS:  NAD83
```



```r
# plot_error_heatmap()
box::reload(plot_maps)
heatmap <- plot_maps$plot_error_heatmaps(syn_hhs_spatial_file="synthpop_output/2503302/syn_hhs_spatial_2
                                          syn_indvs_spatial_file="synthpop_output/2503302/syn_indvs_spatial_2503302-exam
                                          marg_dir="synthpop_data/acs_marginals/2503302/", save_pdf=TRUE)
```

```r
# plot_error_map
map_pumas_subset <- c(2503302)
box::reload(plot_maps)
plot_maps$plot_error_map("Spatial errors in tenure by houshold income distribution",
                        map_pumas_subset, densities,
                        variable="tenur_hhinc",
                        spatial_level="tract",
                        synpop_suffix=synpop_suffix,
                        gray_hh_pop_under=30,
                        gray_indv_pop_under=50,
                        NA_label="too few households",
                        save_pdf=TRUE)
```

```
## [1] 2503302
```




```r
#synthpop/synthpop_output/2503302/syn_hhs_spatial_2503302.csv
```

tmap fixes

tmap

```r
# .libPaths()
# # Re-install the v3 branch directly to renv environment
# renv::purge("tmap")
# renv::install("r-tmap/tmap@v3")
#
# # Snapshot the environment state to update your renv.lock file
# renv::snapshot(prompt = FALSE)
# packageVersion("tmaptools")
```

tmaptools

```r
# # 1. Wipe the current mismatched tmaptools
# renv::purge("tmaptools")
#
# # 2. Install the legacy version of tmaptools that matches tmap v3
# # Version 3.1-1 is the stable companion release for tmap v3
# renv::install("cran/tmaptools@3.1-1")
#
# # 3. Reinstall tmap@v3 just to ensure dependencies hook back together cleanly
# renv::install("r-tmap/tmap@v3", upgrade = "never")
#
# # 4. Lock this working combination into your renv lockfile
# renv::snapshot(prompt = FALSE)
```